3-25-26 (WEEK 5)

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DECOMPOSITION OF IDEALS IN DEDEKIND DOMAINS

PROPOSITION

Let $\mathfrak{a}_1, \mathfrak{a}_2$ nonzero ideals in $\mathcal{O}_K$, then $Nr_{K/\mathbb{Q}}(\mathfrak{a}_1\mathfrak{a}_2) = Nr_{K/\mathbb{Q}}(\mathfrak{a}_1)\, Nr_{K/\mathbb{Q}}(\mathfrak{a}_2)$

Proof

If $\mathfrak{a}_1$ or $\mathfrak{a}_2 = \mathcal{O}_K$, then the statement is trivial.

Consider $\mathfrak{a}_1, \mathfrak{a}_2 \neq \mathcal{O}_K$.

$\because$ $\mathcal{O}_K$: DDK

$\therefore$ $\mathfrak{a}_i := \prod_i \mathfrak{p}_i^{e_i}$. Thus, it suffices to show for $\boxed{\mathfrak{a}_1 : \max}$, $Nr_{K/\mathbb{Q}}(\mathfrak{a}_1\mathfrak{a}_2) = Nr_{K/\mathbb{Q}}(\mathfrak{a}_1)\, Nr_{K/\mathbb{Q}}(\mathfrak{a}_2)$

Notice, $Nr_{K/\mathbb{Q}}(\mathfrak{a}_1\mathfrak{a}_2) = [\mathcal{O}_K : \mathfrak{a}_1\mathfrak{a}_2] = \underbrace{[\mathcal{O}_K : \mathfrak{a}_2]}_{Nr_{K/\mathbb{Q}}(\mathfrak{a}_2)}[\mathfrak{a}_2 : \mathfrak{a}_1\mathfrak{a}_2]$ ($\Rightarrow$ Goal: $[\mathfrak{a}_2 : \mathfrak{a}_1\mathfrak{a}_2] = k$)

$\mathfrak{a}_1\mathfrak{a}_2 \subseteq \mathfrak{a}_2 \subseteq \mathcal{O}_K \Rightarrow \mathfrak{a}_2/\mathfrak{a}_1\mathfrak{a}_2$ is a $\mathcal{O}_K/\mathfrak{a}_1$-module

Need so eval dim is valid

Also, $\mathfrak{a}_1 : \max \Rightarrow \mathcal{O}_K/\mathfrak{a}_1$: field

Claim: $\dim_{\mathcal{K}(\mathfrak{a}_1)} \mathfrak{a}_2/\mathfrak{a}_1\mathfrak{a}_2 = 1$ for $\mathcal{K}(\mathfrak{a}_1) := \mathcal{O}_K/\mathfrak{a}_1$

Proof

If $\dim_{\mathcal{K}(\mathfrak{a}_1)} \mathfrak{a}_2/\mathfrak{a}_1\mathfrak{a}_2 > 1$, then $\exists\, 0 \neq \bar{I} \subsetneq \mathfrak{a}_2/\mathfrak{a}_1\mathfrak{a}_2$: proper subspace over $\mathcal{K}(\mathfrak{a}_1)$

Define: $\mathfrak{a}_2 \xrightarrow{\pi} \mathfrak{a}_2/\mathfrak{a}_1\mathfrak{a}_2$

$\pi^{-1}(\bar{I}) \leftsquigarrow \bar{I}$

As $\bar{I}$: $\mathcal{O}_K/\mathfrak{a}_1$-vector space, thus $\underbrace{\pi^{-1}(\bar{I})}_{0 \neq \pi^{-1}(\bar{I}) \subseteq \mathcal{O}_K}$ is a $\mathcal{O}_K$-module

Thus, $\mathfrak{a}_1\mathfrak{a}_2 \subseteq \pi^{-1}(\bar{I}) \subsetneq \mathfrak{a}_2$

As $\mathfrak{a}_1$: max, by unique fact of DDK, $\pi^{-1}(\bar{I}) = \mathfrak{a}_1\mathfrak{a}_2$ or $\mathfrak{a}_2$ $\rightarrow\!\leftarrow$

$\therefore$ $\dim_{\mathcal{K}(\mathfrak{a}_1)} \mathfrak{a}_1/\mathfrak{a}_1\mathfrak{a}_2 = 1 \Rightarrow \# \mathfrak{a}_1/\mathfrak{a}\mathfrak{a}_2 = \# \mathcal{K}(\mathfrak{a}_1)$ $\square$

LEMMA

(i) A: DDK domain, $(0) \neq I \trianglelefteq A$: proper ideal. Let $P \trianglelefteq A$: max ideal, then $I \subseteq P \Leftrightarrow P \mid I$

Proof: By def $\square$

(ii) $0 \neq \mathfrak{a} \trianglelefteq \mathcal{O}_K$, $n = N_{K/\mathbb{Q}}(\mathfrak{a}) = [\mathcal{O}_K : \mathfrak{a}] \Rightarrow n \in \mathfrak{a} \cap \mathbb{Z}$

EXAMPLE

$K = \mathbb{Q}(\sqrt{d})$, $d \equiv 2, 3 \pmod 4$

Recall: For prime $p \in \mathbb{Z}$,

① $\mathbb{F}_p[x]/(x^2-d) \cong \mathbb{F}_p \times \mathbb{F}_p = \mathbb{F}_p[x]/(x-u) \times \mathbb{F}_p[x]/(x+u)$ where $u \in \mathbb{Z}$, $u^2 \equiv d \pmod p$, $p \neq 2$

② $0 \neq d \notin (\mathbb{F}_p^\times)^2 \Rightarrow \mathbb{F}_p[x]/(x^2-d)$: field

③ $p \mid d \rightsquigarrow (p, \sqrt{d})$: only ideal over p

④ $p = 2$, $p \nmid d \rightsquigarrow (p, 1-\sqrt{d})$: only ideal over p

We can calculate:

① $\Rightarrow (p, u-\sqrt{d}), (p, u+\sqrt{d})$ lying over p

$p\mathcal{O}_K = (p, u-\sqrt{d})^\alpha (p, u+\sqrt{d})^\beta$, $\alpha, \beta \geq 1$ (by Lemma (i))

As $\underset{\substack{\shortparallel \\ p^2}}{Nr_{K/\mathbb{Q}}(p\mathcal{O}_K)} \overset{\text{Prop}}{=} [Nr_{K/\mathbb{Q}}(p, u-\sqrt{d})]^\alpha [Nr_{K/\mathbb{Q}}(p, u+\sqrt{d})]^\beta$, thus $\alpha = \beta = 1$, $Nr_{K/\mathbb{Q}}(p, u-\sqrt{d}) = Nr_{K/\mathbb{Q}}(p, u+\sqrt{d}) = p$

② $\Rightarrow p\mathcal{O}_K$: max

③ $\Rightarrow p\mathcal{O}_K = (p, \sqrt{d})^\alpha$

$Nr_{K/\mathbb{Q}} \Rightarrow p^2 = Nr(p, \sqrt{d})^\alpha = p^\alpha \Rightarrow \alpha = 2$

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EXAMPLE

$d=-5$

Then $(1+2\sqrt{-5}) \trianglelefteq \mathcal{O}_K \overset{Nr_{K/\mathbb{Q}}}{\Longrightarrow} Nr_{K/\mathbb{Q}}(1+2\sqrt{-5}) = N_{K/\mathbb{Q}}(1+2\sqrt{-5}) = 21$

By Lemma (ii), $21 \in P_i$ if $P_i \supseteq (1+2\sqrt{-5})\mathcal{O}_K$

As P_i: max, thus 3 or $7 \in P_i$

Lying over 3: $\underbrace{(3, 1+\sqrt{-5})}_{\subseteq P_3}$, $\underbrace{(3, 1-\sqrt{-5})}_{\subseteq P_3'}$ $(-5 \equiv 1^2 \pmod 3) \Rightarrow \mathbb{F}_p[x]/(x^2-d) \cong \mathbb{F}_p[x]/x-u \times \mathbb{F}_p[x]/x+u \cong \mathbb{F}_p \times \mathbb{F}_p$, $u=1$

Lying over 7: $(7, 3+\sqrt{-5})$, $(7, 3-\sqrt{-5})$ $(-5 \equiv 3^2 \pmod 7)$

$1+2\sqrt{-5} \rightsquigarrow 1+2x$.

$(3, 1+\sqrt{-5}) \rightsquigarrow x \equiv -1 \pmod 3 \Rightarrow 1+2x \not\equiv 0 \pmod 3$ ✗

$(3, 1-\sqrt{-5}) \rightsquigarrow x \equiv 1 \pmod 3 \Rightarrow 1+2x \equiv 0 \pmod 3$ ✓

$(7, 3+\sqrt{-5}) \rightsquigarrow x \equiv -3 \pmod 7 \Rightarrow 1+2x \not\equiv 0 \pmod 7$ ✗

$(7, 3-\sqrt{-5}) \rightsquigarrow x \equiv 3 \pmod 7 \Rightarrow 1+2x \equiv 0 \pmod 7$ ✓

We can directly check $(1+2\sqrt{-5})\mathcal{O}_K = (3, 1-\sqrt{-5})(7, 3-\sqrt{-5})$

(Or compare the behavior of P_3, P_3' and $(1+2\sqrt{-5})\mathcal{O}_K$ in mod 3)

REVISITING FIRST EXAMPLE

① $\alpha = \beta = 1$: $\mathcal{O}_K/B = \mathbb{F}_p = \mathbb{Z}/p$ $\forall B \supseteq p\mathcal{O}_K$

② $[\mathcal{O}_K/p\mathcal{O}_K : \mathbb{Z}/p] = 2$, $\mathcal{O}_K = p\mathcal{O}_K$

③ $p\mathcal{O}_K = (p, \sqrt{d})^2$, $\mathcal{O}_K/(p,\sqrt{d}) = \mathbb{Z}/p$

Let $p\mathcal{O}_K = \prod B_i^{e_i}$ be the factorization in $\mathcal{O}_K$.

①②③ $\Rightarrow$ $[K:\mathbb{Q}] = \sum_i e_i f_i$, $f_i = [\mathcal{O}_K/B_i : \mathbb{Z}/p]$

① $\Leftrightarrow$ $e_1 = e_2 = 1$, $f_1 = f_2 = 1$

② $\Leftrightarrow$ $i=1$, $e_1 = 1$, $f_1 = 2$

③ $\Rightarrow$ $i=1$, $e_1 = 2$, $f_1 = 1$

FIX NOTATION

$\mathcal{O}_K$: DDK Domain

$K := \mathrm{Frac}(\mathcal{O}_K)$

L/K: fin+sep ext

$\mathcal{O}_L$: Int closure of $\mathcal{O}_K$ in L, also a DDK Domain

DEFINITION

Given a max ideal $P \trianglelefteq \mathcal{O}_K$, and $P\mathcal{O}_L = \prod_{i=1}^n B_i^{e_i}$ with B_i: distinct max, $e_i > 1$

We define:

(i) e_i is called the ramification index of B_i over P

(ii) $f_i := [\mathcal{O}_L/B_i : \mathcal{O}_K/p]$ is called the inert degree of B_i over P

$\kappa(B_i) = \mathcal{O}_L/B_i$: residue field of B_i

(iii) p splits completely if $e_i = 1$ and $f_i = 1$ $\forall i$

(iv) p is unramified if $e_i = 1$ $\forall i$. O/w, we say it's ramified

(v) p is totally ramified if $n=1$, $f_1 = 1$

(vi) p is inert if $n=1$, $e_1 = 1$

PROPOSITION

Let $p \trianglelefteq \mathcal{O}_K$ be a max ideal. Write $p\mathcal{O}_L = \prod_{i=1}^n B_i^{e_i}$, $e_i > 0$, B_i: distinct, max

(i) $\mathcal{O}_L/p\mathcal{O}_L$ is an $\mathcal{O}_K/p$-vec space of dim $= [L:K]$

(ii) $[L:K] = \sum_{i=1}^n e_i f_i$

Proof

(i) Consider localization $\Rightarrow$ PID

$S := \mathcal{O}_K \setminus p \Rightarrow S^{-1}\mathcal{O}_K/pS^{-1}\mathcal{O}_K = \mathcal{O}_K/p\mathcal{O}_K$, $S^{-1}\mathcal{O}_L/pS^{-1}\mathcal{O}_L = \mathcal{O}_L/p\mathcal{O}_L$

localization $\Rightarrow$ only one prime exists

By exercise, $S^{-1}\mathcal{O}_K = \mathcal{O}_{K,p}$ is a PID; $K = \mathrm{Frac}(\mathcal{O}_{K,p})$, $S^{-1}\mathcal{O}_L$: f.g. mod $/\mathcal{O}_{K,p} \Rightarrow \mathcal{O}_L$: f.g. $\mathcal{O}_K$-module □

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(ii) By CRT, $\mathcal{O}_L/p\mathcal{O}_L = \prod_{i=1}^{n} \mathcal{O}_L/B_i^{e_i}$ and $\dim_{\kappa(p)} \mathcal{O}_L/p\mathcal{O}_L = [L:K]$ by (i)

It suffices to show $\dim_{\kappa(p)} \mathcal{O}_L/B_i^{e_i} = e_i f_i$.

By def, $f_i := [\mathcal{O}_L/B_i : \kappa(p)]$

(Rmk: A, B, C: abelian groups
$A \xrightarrow{f} B \xrightarrow{g} C$, f, g: group homo
$\mathrm{Im}(f) = \mathrm{Ker}(g) \Rightarrow$ the seq is "exact")

代数学の先生にごめんと言う…

Consider this exact sequence of $\kappa(p)$-vec space. not $\mathcal{O}_L/B$

$0 \to B^i/B^{i+1} \hookrightarrow \mathcal{O}_L/B^{i+1} \xrightarrow{\pi_i} \mathcal{O}_L/B^i \to 0$

Then, $\dim_{\kappa(p)} \mathcal{O}_L/B^{i+1} = \dim_{\kappa(p)} \mathcal{O}_L/B^i + \dim_{\kappa(p)} B^i/B^{i+1}$

has only one max ideal B

Claim: $\mathcal{O}_L/B \curvearrowright B^i/B^{i+1} \subseteq \mathcal{O}_L/B^{i+1}$

Proof: Choose $\alpha \in B \setminus B^2$, then $(\alpha^i)\mathcal{O}_L + B^{i+1} \trianglelefteq \mathcal{O}_L$, but also $B^{i+1} \subsetneq (\alpha^i)\mathcal{O}_L + B^{i+1} \subseteq B^i \Rightarrow (\alpha^i)\mathcal{O}_L + B^{i+1} = B^i$

$\therefore B^i/B^{i+1} = (\bar{\alpha})$, $\bar{\alpha}$: Im of α in B^i/B^{i+1}

$\Rightarrow \dim_{\kappa(p)} B^i/B^{i+1} = 1$

$$\begin{aligned} \mathcal{O}_L/B &\xrightarrow{\sim} B^i/B^{i+1} \\ 1 &\longmapsto \alpha \end{aligned}$$

$\Rightarrow \dim_{\kappa(p)} B^i/B^{i+1} = \dim_{\kappa(p)} \mathcal{O}_L/B = f$

Inductively, $e_i f_i = \dim_{\kappa(p)} \mathcal{O}_L/B_i^{e_i}$ $\square$

COROLLARY

Assume L/K: fin Gal. Then, $[L:K] = efg$, $g = \#$distinct max ideal

LEMMA

Given L/K: fin Gal, max $P \trianglelefteq \mathcal{O}_K$, and $P\mathcal{O}_L = \prod B_i^{e_i}$

Then, $\mathrm{Gal}(L/K)$ acts transitively on B_i

Proof

Suppose not, then $\exists B \in \{B_2, \dots, B_n\}$ s.t. $\sigma(B_1) \neq B$ $\forall \sigma \in \mathrm{Gal}(L/K)$

Choose $x \in \mathcal{O}_L$, s.t. $\begin{cases} x \equiv 0 \mod B \\ x \equiv 1 \mod (\sigma(B_1)) \ \forall \sigma \in \mathrm{Gal}(L/K) \end{cases}$

$\Rightarrow N_{L/K}(x) \in \mathcal{O}_K \cap B = P$

$\therefore \forall \sigma(x) \notin B_1$, $\forall \sigma$, $N_{L/K}(x) = \prod_{\sigma \in \mathrm{Gal}(L/K)} \sigma(x) \notin B_1 \cap \mathcal{O}_K = P$ $\rightarrow\leftarrow$

PROOF OF COROLLARY

$p\mathcal{O}_L = \prod B_i^{e_i} \Rightarrow \sigma(p\mathcal{O}_L) = \prod \sigma(B_i)^{e_i}$

By lemma, $\exists \sigma$, s.t. $\sigma(B_i) = B_1$ $\forall i$

$\therefore e_i = e_1$, $\mathcal{O}_L/B_1 \xrightarrow[\sim]{\sigma} \mathcal{O}_L/\sigma(B_1)$: $\kappa(p)$-vec morphism $\square$